TTARAG: Predict the Retrieval! Test Time Adaptation for Retrieval Augmented Generation

Xin Sun[1], Zhongqi Chen[2], Qiang Liu[1], Shu Wu[1], Bowen Song[2], Weiqiang Wang[2], Zilei Wang[3], Liang Wang[1]

[1] NLPR, MAIS, CASIA [2] Ant Group [3] USTC

ICASSP 2026

GitHub Code

TL;DR

We propose TTARAG — a test-time adaptation method that dynamically updates LLM parameters during inference by learning to predict retrieved content, enabling automatic domain adaptation for RAG systems without any labeled data.

- **Problem:** RAG systems struggle with distribution shifts in specialized domains (medical, finance, etc.), often failing to utilize domain-specific context effectively
- **Self-supervised adaptation:** Split retrieved passages into prefix-suffix pairs; train the model to predict suffix from prefix + query, learning domain-specific language patterns
- **No labeled data required:** Fully unsupervised — adapts using only the retrieved passages available at test time, no source domain access needed
- **Consistent improvements:** Achieves the best results in 19 out of 24 experimental settings across 6 specialized domains with 3 different model architectures
- **Dramatic medical QA gains:** Up to +25.0% on PubMedQA, +19.4% on BioASQ — significantly outperforming CoT, ICL, and pretrained RAG models
- **Computationally efficient:** 40% faster than Chain-of-Thought reasoning while achieving superior performance

Framework Overview

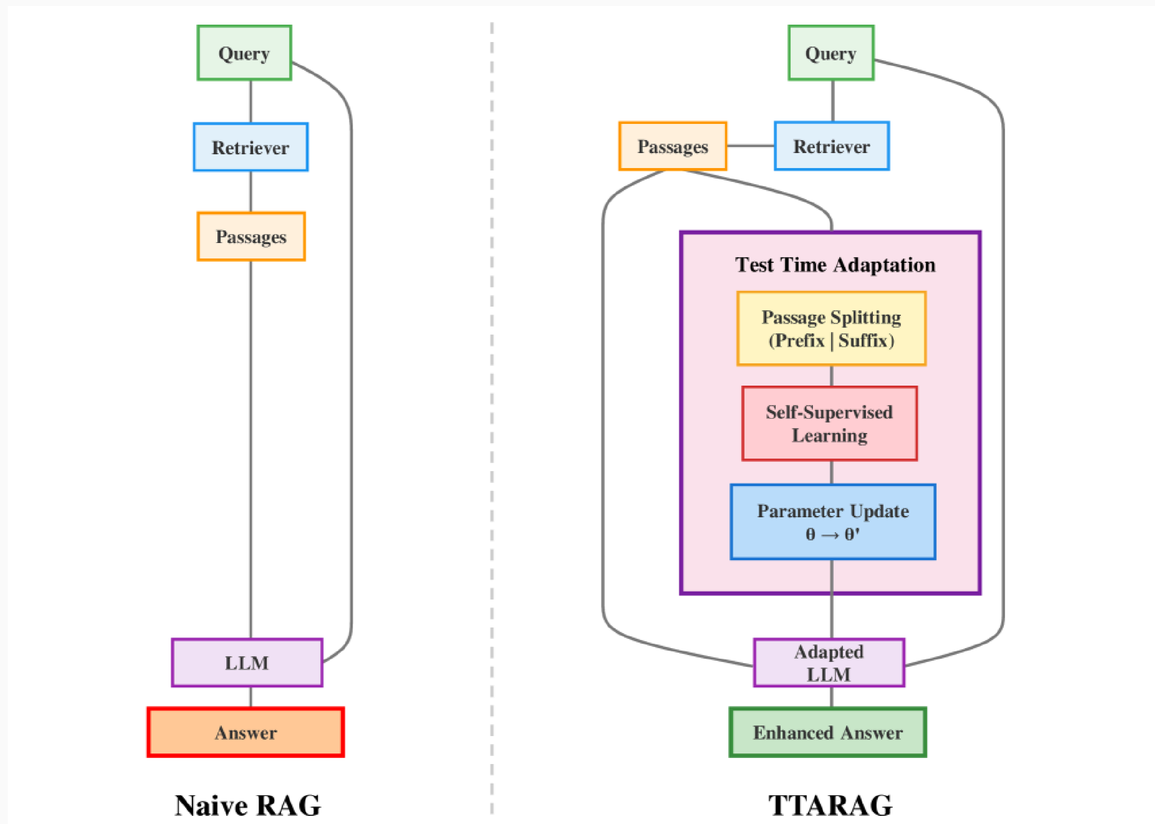

Figure 1. Standard RAG (left) directly uses retrieved passages. TTARAG (right) introduces a test-time adaptation module: passages are split into prefix-suffix pairs → self-supervised learning predicts suffixes → model parameters are updated before final answer generation.

Methodology

Key insight: By predicting suffix from prefix, the model is encouraged to understand contextual relationships and domain-specific language patterns within retrieved documents, aligning its internal representations with the target domain.

Step 1 — Self-Supervised Adaptation Objective:

$$\mathcal{L}_{adapt} = -\sum_{i=1}^{k} \log P(p_i^{suffix} \mid p_i^{prefix}, q;\ \theta)$$

- **Length Filtering:** Passages shorter than a configured minimum length threshold are filtered out to ensure sufficient context for meaningful adaptation
- **Passage Splitting:** Two-tier strategy — (1) split at natural linguistic boundaries (periods, commas, semicolons); (2) fallback to midpoint division ensuring ≥3 words per segment
- **Parameter Update:** Reset to pretrained state per query; AdamW optimizer with gradient accumulation over 2 steps and gradient clipping for stable training. Only 3 prefix-suffix pairs needed

Step 2 — Response Generation with adapted θ':

$$y = \arg\max_y P(y \mid q, \{p_1, \ldots, p_k\};\ \theta')$$

The adapted model generates the final response using the domain-aligned parameters, leveraging both the retrieved passages and the newly acquired domain knowledge.

Main Results — Accuracy (%)

Model	Finance	Sports	Music	Movie	Open	Overall	BioASQ	PubMed
Llama-3.1-8b-it								
naive-rag	17.4	27.6	34.9	31.3	42.4	29.8	55.6	46.6
+CoT	17.9	30.2	37.6	31.5	45.8	31.6	54.6	50.8
+ICL	16.1	24.8	33.5	29.4	40.4	28.0	49.8	53.6
TTARAG	**20.1**	29.5	**37.7**	**34.6**	41.5	**31.9**	**75.0**	**57.4**
Δ vs naive	+2.7	+1.9	+2.8	+3.3	-0.9	+2.1	+19.4	+10.8
Llama-2-7b-chat								
naive-rag	14.7	23.2	36.5	30.4	39.2	27.8	54.1	47.6
TTARAG	**16.4**	25.8	**40.7**	**33.8**	41.1	**30.5**	**71.8**	**54.0**
Δ vs naive	+1.7	+2.6	+4.2	+3.4	+1.9	+2.7	+17.7	+6.4
ChatGLM-3-6b								
naive-rag	9.8	18.7	31.4	22.4	33.4	22.0	51.4	19.8
TTARAG	**14.0**	**22.1**	**33.5**	25.5	**38.1**	**25.7**	**58.4**	**44.8**
Δ vs naive	+4.2	+3.4	+2.1	+3.1	+4.7	+3.7	+7.0	+25.0

vs. Pretrained RAG Models

All models use Llama-2-7b-chat backbone

Model	Finance	Sports	Music	Movie	Open	Overall	BioASQ	PubMed
Naive-RAG	14.7	23.2	36.5	30.4	39.2	27.8	54.1	47.6
Ret-Robust	14.6	20.6	33.2	32.4	33.5	26.1	24.7	28.4
RAAT	13.4	18.1	28.6	25.2	31.7	22.7	64.9	46.6
Self-RAG	11.4	19.8	22.5	20.9	26.7	19.8	57.1	43.4
TTARAG	**16.4**	**25.8**	**40.7**	**33.8**	**41.1**	**30.5**	**71.8**	**54.0**

Ablation: Segmentation

Prefix-suffix segmentation consistently outperforms whole-passage next-token prediction: +1.1% (Llama-3.1-8b), +0.4% (Llama-2-7b), +0.7% (ChatGLM-3-6b)

- The front-to-back prediction task better aligns with natural language understanding compared to token-by-token prediction
- Higher-capacity models benefit more from structured adaptation (+1.1% for Llama-3.1 vs +0.4% for Llama-2)

Hyperparameter Analysis

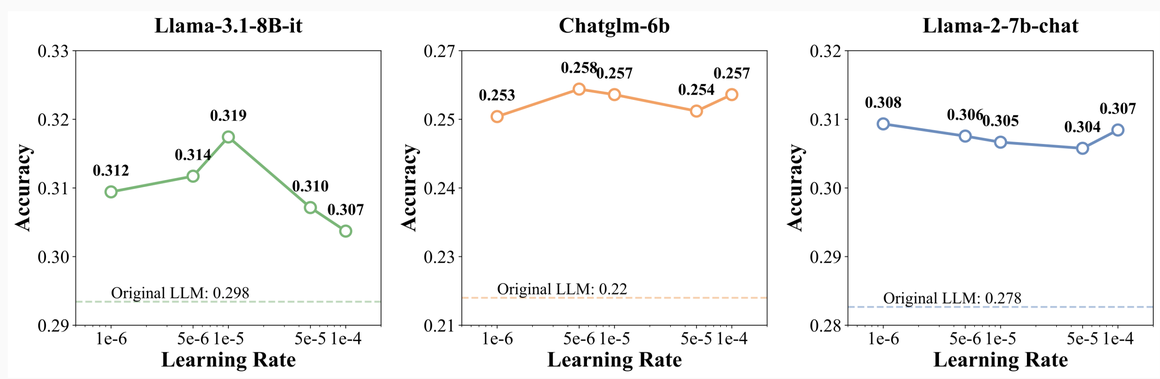

Figure 2. Accuracy vs. Learning Rate — optimal range: 1e-6 to 1e-5

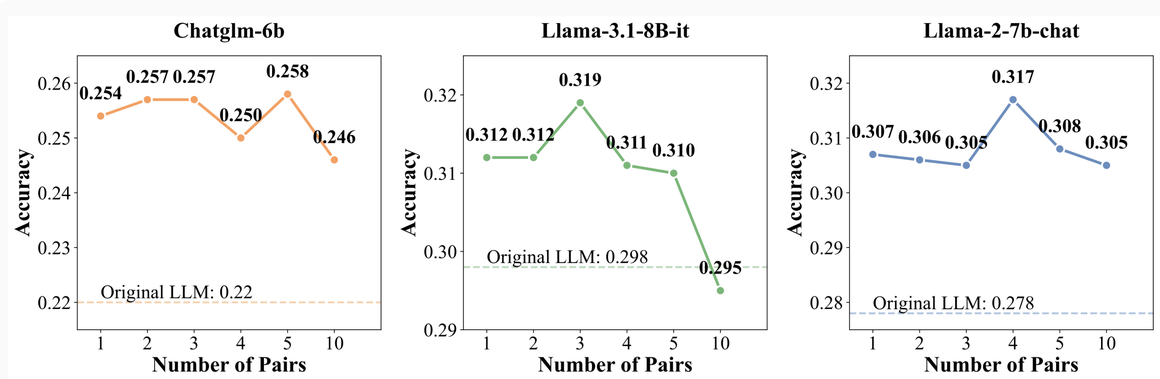

Figure 3. Accuracy vs. Number of Adaptation Pairs — 3-5 pairs yield best results

Computational Efficiency

Metric	1 pair	3 pairs	5 pairs	CoT	Naive
Total (s)	4,740	6,621	7,023	11,688	961
Avg (s/query)	1.75	2.45	2.60	4.32	0.36

Conclusion

- **TTARAG** enables fully unsupervised domain adaptation for RAG at inference time
- Simple prefix-suffix prediction provides effective self-supervised signals — no labeled data needed
- Up to **+25%** improvement on specialized domains; **40% faster** than Chain-of-Thought
- Test-time adaptation is a promising direction for domain-specific RAG systems

Code: github.com/sunxin000/TTARAG
Paper: arXiv:2601.11443
Contact: xin.sun@cripic.ia.ac.cn